A common-burden sensitivity calculation

Native reconstruction budgets require native burden and loss-frequency series. When only an annual OH removal flux Q_j is supplied, introduce an explicitly conditional reference burden $m_j^* = cX_j$, in which X_j is NOAA annual mean surface methane and $c = 2.77$ Tg ppb^{-1} is He's conversion factor. Define

$$F^*_{\mathrm{OH,j}} = -m_0^* \left(\frac{Q_j}{m_j^*} - \frac{Q_0}{m_0^*} \right) = Q_0 - Q_j \frac{X_0}{X_j}. \tag{1}$$

This is He's baseline-burden forcing evaluated with a common observational proxy. It is not a native model diagnostic: surface and whole-atmosphere abundances and their temporal evolution can differ. A constant multiplicative error in c cancels; a time-varying representativeness error does not. For observed growth $G_j^* = cg_j$, the source-plus-OH forcing required under fixed soil and other chemical loss frequencies is

$$F^*_{\mathrm{req,j}} = c(g_j - g_0) + \frac{c(X_j - X_0)}{\tau_0}. \tag{2}$$

We examine chemical lifetimes $\tau_0 = 9, 10, 11, 12$ years as a sensitivity range, not as posterior estimates. It brackets the approximately 11-year OH lifetimes in He and Qu and allows for additional chemical removal. Soil uptake is excluded from τ_0, following He. Annual mean burden is an approximation to the burden in an annual integrated budget. We therefore do not fit these scenarios to the observed growth or treat discrepancies as a new flux inversion.